

SKILL WEEK9

TASK1: To Implement how to List Active Jobs, Display Status Information, Retrieve Job IDs, Format Output, Update State Information, Test Job Listings.

```
rishill10@ASUS16X: ~/Skill9
rishill10@ASUS16X:~$ mkdir Skill9
cd Skill9
rishill10@ASUS16X:~/Skill9$ sleep 100 &
[1] 545
rishill10@ASUS16X:~/Skill9$ jobs
[1]+  Running                  sleep 100 &
rishill10@ASUS16X:~/Skill9$ jobs -l
[1]+  545 Running                  sleep 100 &
rishill10@ASUS16X:~/Skill9$ jobs -p
545
rishill10@ASUS16X:~/Skill9$ jobs -x echo %1
545
rishill10@ASUS16X:~/Skill9$ bg %1
-bash: bg: job 1 already in background
rishill10@ASUS16X:~/Skill9$ jobs
fg %1
[1]+  Running                  sleep 100 &
sleep 100
^C
rishill10@ASUS16X:~/Skill9$ |
```

TASK2: To design how to Identify Target Job, Transfer Terminal Control, Resume Processes, Wait for Completion, Update Job State, Test Foreground Switching.

```
rishill10@ASUS16X: ~/Skill9
rishill10@ASUS16X:~/Skill9$ sleep 200
^Z
[1]+  Stopped                  sleep 200
rishill10@ASUS16X:~/Skill9$ jobs
[1]+  Stopped                  sleep 200
rishill10@ASUS16X:~/Skill9$ bg %1
[1]+  sleep 200 &
rishill10@ASUS16X:~/Skill9$ jobs
[1]+  Running                  sleep 200 &
rishill10@ASUS16X:~/Skill9$ fg %1
sleep 200
^C
rishill10@ASUS16X:~/Skill9$ |
```

TASK3: To Analyze Signal Delivery, Understand Signal Masks, Review Process Groups, Explore Terminal Signals, Test Signal Behavior

```
GNU nano 8.7.1 signal.c *
#include <stdio.h>
#include <signal.h>
#include <unistd.h>

void handle_signal(int sig)
{
    printf("\nSignal received: %d\n", sig);
}

int main()
{
    signal(SIGINT, handle_signal);
    signal(SIGTERM, handle_signal);

    printf("Process ID: %d\n", getpid());
    printf("Press Ctrl+C to send SIGINT\n");

    while (1)
    {
        printf("Running...\n");
        sleep(2);
    }
    return 0;
}
```

[illegible]

```
rishill10@ASUS16X: ~/Skill9  ×  Ubuntu  ×  +  ▾  -  □  ×

rishill10@ASUS16X:~$ kill -TERM 727
-bash: kill: (727) - No such process
rishill10@ASUS16X:~$ kill -15 727
-bash: kill: (727) - No such process
rishill10@ASUS16X:~$ kill -9 727
-bash: kill: (727) - No such process
rishill10@ASUS16X:~$ \kill -l
 1) SIGHUP      2) SIGINT      3) SIGQUIT     4) SIGILL      5) SIGTRAP
 6) SIGABRT     7) SIGBUS     8) SIGFPE     9) SIGKILL     10) SIGUSR1
11) SIGSEGV    12) SIGUSR2   13) SIGPIPE   14) SIGALRM    15) SIGTERM
16) SIGSTKFLT  17) SIGCHLD   18) SIGCONT    19) SIGSTOP    20) SIGTSTP
21) SIGTTIN    22) SIGTTOU   23) SIGURG     24) SIGXCPU    25) SIGXFSZ
26) SIGVTALRM  27) SIGPROF   28) SIGWINCH   29) SIGIO       30) SIGPWR
31) SIGSYS     34) SIGRTMIN  35) SIGRTMIN+1 36) SIGRTMIN+2 37) SIGRTMIN
+3
38) SIGRTMIN+4 39) SIGRTMIN+5 40) SIGRTMIN+6 41) SIGRTMIN+7 42) SIGRTMIN
+8
43) SIGRTMIN+9 44) SIGRTMIN+10 45) SIGRTMIN+11 46) SIGRTMIN+12 47) SIGRTMIN
+13
48) SIGRTMIN+14 49) SIGRTMIN+15 50) SIGRTMAX-14 51) SIGRTMAX-13 52) SIGRTMAX
-12
53) SIGRTMAX-11 54) SIGRTMAX-10 55) SIGRTMAX-9 56) SIGRTMAX-8 57) SIGRTMAX
-7
58) SIGRTMAX-6 59) SIGRTMAX-5 60) SIGRTMAX-4 61) SIGRTMAX-3 62) SIGRTMAX
-2
63) SIGRTMAX-1 64) SIGRTMAX
rishill10@ASUS16X:~$ ps aux | grep signal
root      281  0.0  0.4 123456 32624 ?        Ssl  02:56   0:00 /usr/bin/
python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-fo
r-signal
rishill+  815  0.0  0.0   2768  1884 pts/0    S+   03:06   0:00 ./signal
rishill+  914  0.0  0.0   4128  2384 pts/2    S+   03:09   0:00 grep --co
lor=auto signal
rishill10@ASUS16X:~$ cat /proc/727/status | grep Sig
cat: /proc/727/status: No such file or directory
rishill10@ASUS16X:~$
```

TASK4: To implement how to resume Stopped Jobs, Send Continue Signals, Update Job Status, Maintain Job Table, Verify Execution, Test Background Recovery.

```
rishill10@ASUS16X: ~/Skill9
rishill10@ASUS16X:~/Skill9$ cd Skill9
rishill10@ASUS16X:~/Skill9$ nano resume_jobs.c
rishill10@ASUS16X:~/Skill9$ gcc resume_jobs.c -o resume_jobs
./resume_jobs
Session 9 - Resume Stopped Jobs
Commands: run <command>, jobs, resume <job_id>, exit
myshell> run sleep 20
sleep: missing operand
Try 'sleep --help' for more information.
Job [1] started and stopped. PID: 1094
myshell> jobs

Job Table:
-----
Job ID  PID    Status    Command
-----
[1]     1094    Completed  sleep
-----

myshell> resume 1
Job [1] is not in Stopped state (Current: Completed).
myshell> jobs

Job Table:
-----
Job ID  PID    Status    Command
-----
[1]     1094    Completed  sleep
-----

myshell> exit
Exiting...
rishill10@ASUS16X:~/Skill9$ |
```